WEEKLY LESSON PLAN · 2026-27

Manufacturing Engineering Technology I

TEKS §127.813 (working draft) · CTE Level 2 · Grade 10 · Mr. Gavaldon · Da Vinci School
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SEMESTER 1
Week 2

MONDAY Day 6 - Launch: Engineering Firms + Pick Your Problem

Aug 17

OBJECTIVE
Students will form engineering firms, choose a real problem, and define why it matters (Design Brief p.1).

TEKS
§127.813 (draft, verify strand) measurement, tools & precision in manufacturing

ELPS / EB SUPPORT
ELPS/EB supports at-level: pre-teach accuracy/precision/tolerance with a picture card (PV, DP); model the vernier reading step-by-step and repeat slowly (RS, VC); hands-on measuring stations with a partner and native-language number talk (PN, CD); wait time on the classify-the-data exit question (WT).

AGENDA · 45 MIN
- 6′ Bell-work (format + example): name a badly-designed everyday thing
- 10′ Launch: the firm mission + week map
- 8′ Worked example: a firm end-to-end
- 6′ Randomize firms + assign 4 roles
- 15′ Pick your problem + start Design Brief page 1

SLIDES / ACTIVITY
Engineering Firms – Launch

NOTEBOOK
Record your station measurements. Define precision vs. tolerance in your own words.

TURN IN
Engineering Firms - Design Brief packet, page 1 (firm, roles, problem, why)

TUESDAY Day 7 - Define the Problem

Aug 18

OBJECTIVE
Students will define their problem with a stakeholder, evidence, testable requirements, and a success statement (Design Brief p.2).

TEKS
§127.813 (draft, verify strand) applied mathematics & measurement in manufacturing

ELPS / EB SUPPORT
ELPS/EB supports at-level: keep a conversion reference card (1 in = 25.4 mm) and worked-example frames (CD, RS); pre-teach nominal / tolerance / limits / stack-up (PV); dual-language number talk with a partner (PN); the mm = 25.4 x in graph is a visual anchor for the relationship (VC, DP); extra time on the stack-up and fit problems (ET).

AGENDA · 45 MIN
- 6′ Bell-work: who has it + how you know
- 12′ Define like an engineer: who, evidence, requirements, success
- 10′ Worked example + vocab
- 17′ Write 3-5 testable requirements - Design Brief page 2

SLIDES / ACTIVITY
Engineering Firms – Define It

NOTEBOOK
Keep the conversion table. Work 2 example problems with all steps shown.

TURN IN
Engineering Firms - Design Brief packet, page 2 (definition + requirements)

WEDNESDAY Day 8 - Design the Solution + AI Render

Aug 19

OBJECTIVE
Students will sketch multiple solutions, choose one against their requirements, and generate an AI product image (Design Brief p.3).

TEKS
`§127.813 (draft, verify strand) parametric CAD for manufacturable parts`

ELPS / EB SUPPORT
ELPS/EB supports at-level: build steps mirrored on-screen and posted as Classroom screenshots (VC, DP); pre-teach parameter / expression / constraint / fully-constrained (PV); rephrase and slow each build step (RS); finishers pair with peers who are stuck (PN); extra time to reach a fully-constrained sketch (ET).

AGENDA · 45 MIN
- **6′** Bell-work: why sketch more than one idea
- **10′** Sketch 3 solutions + choose with a reason/trade-off
- **10′** Write the product-image prompt + generate
- **19′** Finish Design Brief page 3 (sketches, choice, image)

SLIDES / ACTIVITY
Engineering Firms – Design It

NOTEBOOK
Write the 3 parameters you set (name = value) and what each controls on the part.

TURN IN
Engineering Firms - Design Brief packet, page 3 (design + AI image)

THURSDAY Day 9 - Build & Drill the Pitch

Aug 20

OBJECTIVE
Students will build a 5-beat pitch deck with AI and prepare to defend it under questioning (Design Brief p.4).

TEKS
`§127.813 (draft, verify strand) apply parametric CAD to a defined part; documentation`

ELPS / EB SUPPORT
ELPS/EB supports at-level: the part-planning sheet organizes the math before CAD (CD, DP); a reference model and a dimensioned sketch are shown on screen (VC); pre-teach clearance hole / edge distance / datum / fastener designation (PV); partner talk in native language while planning (PN); extra time to finish the sketch (ET).

AGENDA · 45 MIN
- **6′** Bell-work: what makes a pitch believable
- **10′** The 5 beats + AI-build the deck from your packet
- **12′** Drill the defense: assign who owns each question
- **17′** Finish deck + one full practice run - Design Brief page 4

SLIDES / ACTIVITY
Engineering Firms – Build the Pitch

NOTEBOOK
Record your plate dimensions and hole sizes. Sketch the plate with dimensions labeled.

TURN IN
Engineering Firms - Design Brief packet, page 4 (pitch deck + Q&A prep)

FRIDAY

Day 10 - Design Review (Capstone)

Aug 21

OBJECTIVE

Student firms will pitch their product and defend it under Q&A from a review panel, scored on a shared rubric.

TEKS

§127.813 (draft, verify strand) industry certification pathway; collaboration

ELPS / EB SUPPORT

ELPS/EB supports at-level: the certification path is shown as a simple roadmap graphic (VC, DP); pre-teach certification / domain / rubric / performance-based (PV); peer-review pairs an EB student with a helper and native-language discussion is welcome (PN); rephrase and slow the test-strategy tips (RS); wait time on the self-check (WT).

AGENDA · 45 MIN

- 6′ Bell-work: what a great audience does
- 6′ Format + rubric (how you are scored)
- 33′ Design Review: 3-min pitch + 2-min defense per firm
- 5′ Exit ticket + Week 3 (30 Days of Fusion) tease

SLIDES / ACTIVITY

Engineering Firms – Design Review

NOTEBOOK

List the certification domains. Note where you are on your plate and one thing to finish Monday.

TURN IN

Engineering Firms - Design Review rubric (25 pts) + reflection